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# $G_2$ Symmetry Breaking in the Universal Somatic Field: The Biological Emotional Attractor and Geometric Ideal

$W = W_{\{G_2\}} + \delta W$ : Decomposing the BRECVEMA Coupling Matrix

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## Abstract

The BRECVEMA coupling matrix  $W_8$  of the Universal Somatic Field decomposes uniquely as  $W_8 = W_{G_2} + \delta W$ , where  $W_{G_2} = \frac{6}{5}I_8$  is the  $G_2$ -invariant component and  $\delta W$  is a traceless symmetry-breaking term. The  $G_2$ -symmetric limit  $W_{G_2}$  corresponds to perfectly balanced emotional processing — all eight BRECVEMA mechanisms equally coupled, no directional anisotropy. The empirical biological matrix  $W_8$  (calibrated from Juslin 2019) is 48.4% symmetry-broken from this ideal:  $\|\delta W\|_F / \|W_8\|_F = 0.484$ . The symmetry-breaking modes are traceless (their eigenvalues sum to zero) and correspond to specific emotional dynamics: strong positive anisotropy in the VI-EM and ME-AJ couplings; negative anisotropy in the BS-AJ channel (stress suppresses aesthetics). Therapeutic intervention reduces  $\|\delta W\|_F$ , driving the system toward the  $G_2$  attractor. This provides the first quantitative connection between a  $G_2$ -symmetric compactification interpretation and the biological coupling matrix. It resolves the algebraic  $8 \rightarrow 7$  reduction in the coupling decomposition; it does not derive a  $G_2$ -holonomy metric for  $X_7$ .



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# 1

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## Introduction: The 8→7 Dimension Question

*Dark Matter as the Spatial Vacuum of the Universal Somatic Field* (Johnson, 2026) proposes that dark matter is the vacuum energy of the three non-compact spatial dimensions of the USF and derived the cosmological energy budget from the dimensional partition  $11 = 7 + 3 + 1$ . The compact sector In an M-theory compactification interpretation, a compact seven-manifold may carry  $G_2$  holonomy. The current USF formal core proves only a well-defined flat 7D product for  $X_7$ . But the biological emotional field is 8-dimensional (BRECHEMA, eight mechanisms). How does its algebraic structure relate to a seven-dimensional compact-sector interpretation?

This paper resolves the question. The 8D BRECHEMA field  $W_8$  decomposes as the  $G_2$ -invariant part plus a traceless symmetry-breaking term. The  $G_2$ -invariant part is exactly  $\frac{6}{5}I_8$  — a diagonal matrix. The remaining 7 off-diagonal degrees of freedom constitute the symmetry-breaking  $\delta W$ , which lives in the 7D adjoint representation of  $G_2$ . The biological emotional system operates on the 8D field, while its traceless symmetry-breaking sector is seven-dimensional. This supplies an algebraic compatibility result for a 7D compact-sector interpretation; it does not by itself prove the metric geometry or holonomy of  $X_7$ .

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## 2

### The $G_2$ -Symmetric Limit

The USF coupling matrix  $W_8$  acts on the 8D BRECVEMA state space  $\psi = (\psi_0, \dots, \psi_7)$  where the indices correspond to the eight mechanisms: BrainStem (BS), Rhythmic Entrainment (RE), Evaluative Conditioning (EC), Contagion (CO), Visual Imagery (VI), Episodic Memory (EM), Musical Expectancy (ME), Aesthetic Judgement (AJ).

**Definition.** A matrix  $W$  acting on  $\mathbb{R}^8$  is  $G_2$ -invariant if it commutes with all  $G_2$  transformations. By Schur's lemma, since  $\mathbb{R}^8$  decomposes under  $G_2$  as  $\mathbb{R}^1 \oplus \mathbb{R}^7$  (real part  $\oplus$  imaginary octonions), a  $G_2$ -invariant matrix must be block-diagonal:  $W_{G_2} = \lambda_0 P_0 + \lambda_1 P_1$ , where  $P_0, P_1$  are projections onto the two invariant subspaces.

For the USF, the self-coupling sets  $\lambda_0 = \lambda_1 = \frac{6}{5}$  (the diagonal of  $W_8$ ), giving:

$$W_{G_2} = \frac{6}{5} I_8$$

This is the  $G_2$ -symmetric attractor: all eight mechanisms are equally coupled, no mechanism is privileged. In the  $G_2$ -symmetric limit, the emotional field has maximal symmetry — no directional anisotropy, no preferred emotional mode.

## 3

### The Decomposition of $W_8$

The empirical matrix  $W_8$  (calibrated from Juslin 2019, Table 22.3) has diagonal entries all equal to  $\frac{6}{5}$  and non-zero off-diagonal entries:

| Coupling    | Value   |
|-------------|---------|
| $W_{BS,EC}$ | $+3/10$ |



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| Coupling    | Value              |
|-------------|--------------------|
| $W_{BS,CO}$ | $+2/5$             |
| $W_{RE,CO}$ | $+1/2$             |
| $W_{EC,CO}$ | $+2/5$             |
| $W_{VI,EM}$ | $+3/5$             |
| $W_{ME,AJ}$ | $+7/10$            |
| $W_{BS,AJ}$ | $-2/5$ (negative)  |
| $W_{EC,VI}$ | $-3/10$ (negative) |

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The unique decomposition  $W_8 = W_{G_2} + \delta W$  gives:

$$\delta W = W_8 - \frac{6}{5}I_8$$

$\delta W$  is traceless by construction:  $\text{tr}(\delta W) = \text{tr}(W_8) - 8 \cdot \frac{6}{5} = \frac{48}{5} - \frac{48}{5} = 0$ .

**Key numerical results:**

$$\|\delta W\|_F = 1.876, \quad \|W_8\|_F = 3.877$$

$$\frac{\|\delta W\|_F}{\|W_8\|_F} = 0.484 \quad (48.4\% \text{ symmetry broken})$$

The eigenvalues of  $\delta W$  (sorted):  $+0.984, +0.718, +0.591, +0.113, -0.226, -0.585, -0.742, -$

Their sum is exactly zero (tracelessness). The spectrum is non-degenerate: biological emotional processing is not  $G_2$ -symmetric at any sub-eigenspace level.

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## 4

## Physical Interpretation of the Symmetry-Breaking Modes

The traceless matrix  $\delta W$  encodes the biological anisotropies of emotional processing. Its non-zero entries correspond to:

**Positive anisotropy** (stronger coupling than the  $G_2$  ideal): -  $\delta W_{ME,AJ} = +0.7$ : Musical expectancy strongly drives aesthetic judgment — the strongest anisotropy in the biological system -  $\delta W_{VI,EM} = +0.6$ : Visual imagery and episodic memory are tightly coupled -  $\delta W_{RE,CO} = +0.5$ : Rhythmic entrainment drives emotional contagion -  $\delta W_{BS,CO} = +0.4$ ,  $\delta W_{EC,CO} = +0.4$ : Arousal and conditioning both activate social contagion

**Negative anisotropy** (weaker coupling than the  $G_2$  ideal; anti-correlation): -  $\delta W_{BS,AJ} = -0.4$ : BrainStem arousal and Aesthetic Judgement are *anti-correlated* in the biological system — when physiological arousal is high, aesthetic appreciation is suppressed. This matches the known psychophysiology of stress and flow states. -  $\delta W_{EC,VI} = -0.3$ : Evaluative conditioning and visual imagery are anti-correlated — conditioned fear suppresses imagery (consistent with PTSD phenomenology)

**The  $G_2$  interpretation:** The positive anisotropies represent the biological “short-cuts” — emotional couplings stronger than the symmetric ideal. The negative anisotropies represent the biological “blockers” — couplings weaker than symmetry would predict.

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5

Therapeutic Trajectory: Reducing  $\delta W$

The decomposition suggests a precise model of the therapeutic process:

**Healthy processing** corresponds to  $\|\delta W\|_F \rightarrow 0$  — the emotional coupling approaching the  $G_2$ -symmetric ideal. Each coupling relaxes toward  $\frac{6}{5}$ : the strongest couplings weaken, the weakest strengthen, the negative couplings (BS–AJ, EC–VI) return to zero.

**Trauma** corresponds to a large  $\|\delta W\|_F$  with specific anisotropies amplified: deep trauma strengthens the BS–AJ anti-correlation (arousal blocks aesthetic experience) and the EC–VI anti-correlation (conditioned responses block visual processing).

**The somatic invariant:**  $\text{tr}(\delta W) = 0$  is preserved throughout. This is the conservation law: the total energy of the symmetry-breaking modes is zero. No therapeutic intervention can add or remove total  $\delta W$  energy — it can only redistribute it. The goal of therapy is to drive  $\delta W$  toward a uniform distribution across all modes (which by tracelessness approaches zero entry-by-entry as the system approaches the  $G_2$  attractor).

6

Formal Status

| Statement                               | Lean location            | Status                                              |
|-----------------------------------------|--------------------------|-----------------------------------------------------|
| $W_{G_2} = (6/5)I_8$ defined            | BRECVEMAVariational.lean | proved (native_decide)                              |
| $\delta W = W_8 - W_{G_2}$ traceless    | BRECVEMAVariational.lean | proved (norm_num)                                   |
| $G_2$ -invariant matrix = $\lambda I_n$ | Schur’s lemma            | interpretation requires<br>Lie-theory formalisation |
| moduli_space_is_G2_homotopy             | BRECVEMAVariational.lean | open formalisation (gauge<br>constraint derivation) |

The numerical decomposition ( $\|\delta W\|_F/\|W_8\|_F = 0.484$ ) is computed exactly using the rational matrix entries; Python floating-point is used only for display. The tracelessness of  $\delta W$  is an exact rational identity.

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## 7

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### Conclusion

The 8-dimensional BRECVEMA coupling matrix  $W_8$  decomposes uniquely as:

$$W_8 = \frac{6}{5}I_8 + \delta W, \quad \text{tr}(\delta W) = 0, \quad \|\delta W\|_F/\|W_8\|_F = 0.484$$

The  $G_2$ -symmetric component  $\frac{6}{5}I_8$  is the mathematical ideal of balanced emotional processing. The traceless symmetry-breaking  $\delta W$  encodes the biological anisotropies: the ME–AJ and VI–EM couplings are the dominant positive anisotropies; the BS–AJ anti-correlation (stress suppresses aesthetics) is the dominant negative anisotropy. Therapeutic progress corresponds to  $\|\delta W\|_F \rightarrow 0$  while  $\text{tr}(\delta W) = 0$  is conserved.

The  $8 \rightarrow 7$  algebraic reduction is resolved: the 8D biological field has a seven-dimensional traceless symmetry-breaking sector, compatible with the 7D compact-sector interpretation used by the companion cosmological-constant and spatial-vacuum papers. Deriving a compact  $G_2$ -holonomy metric remains a separate open geometric problem.

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### References

Johnson, A. (2026). *Dark matter as the spatial vacuum of the universal somatic field*:  $\Omega_{\text{DM}} = 3/11$ .